TRUSTIVA
Efavirenz, Emtricitabine
and Tenofovir Disoproxil
Fumarate Tablets
600 mg/200 mg/300 mg
2087151

TRUSTIVA

Efavirenz, Emtricitabine and Tenofovir Disoproxil Fumarate Tablets
600 mg / 200 mg / 300 mg

Name of Product
Trustiva Tablet

Name and Strength of Active Ingredient
Each film-coated tablet contains:
600 mg of Efavirenz, 200 mg of Emtricitabine and 300 mg of Tenofovir Disoproxil Fumarate is equivalent to 245 mg of Tenofovir Disoproxil.

Dosage form
Oral, fixed-dose tablet

Product Description
Pink colored, capsule shaped, film coated tablets debossed with 'H' on one side and '128' on the other side.

Pharmacodynamics/Pharmacokinetics
Pharmacodynamics
Cardiac Electrophysiology:
The effect of TRUSTIVA on the QTc interval was evaluated in an open-label, positive and placebo controlled, fixed single sequence 3-period, 3-treatment crossover QT study in 58 healthy subjects enriched for CYP2B6 polymorphisms. The mean Cmax of efavirenz in subjects with CYP2B6 *6/*6 genotype following the administration of 600 mg daily dose for 14 days was 2.25-fold the mean Cmax observed in subjects with CYP2B6 *1/*1 genotype. A positive relationship between efavirenz concentration and QTc prolongation was observed. Based on the concentration-QTc relationship, the mean QTc prolongation and its upper bound 90% confidence interval are 8.7 ms and 11.3 ms in subjects with CYP2B6*6/*6 genotype following the administration of 600 mg daily dose for 14 days. (see Section Warnings and Precautions & Section Interaction with Other Medicaments).
Efavirenz
Efavirenz is a non-nucleoside reverse transcriptase (RT) inhibitor of HIV-1. Efavirenz activity is mediated predominantly by the noncompetitive inhibition of HIV-1 reverse transcriptase (RT). HIV-2 RT and human cellular DNA polymerases, α, β, γ and δ are not inhibited by efavirenz.

Emtricitabine
Emtricitabine, a synthetic nucleoside analog of cytidine, is phosphorylated by cellular enzymes to form emtricitabine 5'-triphosphate. Emtricitabine 5'-triphosphate inhibits the activity of the HIV-1 RT by competing with the natural substrate deoxycytidine 5'-triphosphate, and by being incorporated into nascent viral DNA which results in chain termination. Emtricitabine 5'- triphosphate is a weak inhibitor of mammalian DNA polymerase α, β, Є and mitochondrial DNA polymerase, γ.

Tenofovir DF
Tenofovir DF is an acyclic nucleoside phosphonate diester analog of adenosine monophosphate. Tenofovir DF requires initial diester hydrolysis for conversion to tenofovir and subsequent phosphorylations by cellular enzymes to form tenofovir diphosphate. Tenofovir diphosphate inhibits the activity of HIV-1 RT by competing with the natural substrate, deoxyadenosine 5'-triphosphate, and after incorporation into DNA, by DNA chain termination. Tenofovir diphosphate is a weak inhibitor of mammalian DNA polymerases α, β, and mitochondrial DNA polymerase, γ.

Pharmacokinetics
Pharmocokinetics in Adults
The geometric mean ratios of the coformulation to the single agents combination for C_{max}, AUC_{0-T} and AUC0-∞ ranged from 0.89 to 1.0, with 90% CI ranging 84 – 108%, indicating that the two comparators were bioequivalent for all three components, based on these pharmacokinetic parameters (bioequivalence range is 80 – 125%).

The maximum concentration of efavirenz occurs at 2 – 4 h postdose, and are 2.5 – 4.0 mg/l, and minimum concentrations are 1.0 – 1.7 mg/l. Oral clearance is 10 – 16 l/h, with apparent volume of distribution of 350 – 550 l. Efavirenz is 99.5 – 99.75% protein bound, mainly to albumin. AUC at steady-state ranges from 35 – 60 mg/h/l. CYP2B6 and CYP3A4 are the major isozymes that metabolize efavirenz, and efavirenz induces CYP450 enzymes, thereby inducing its own metabolism. A G516T *CYP2B6* gene substitution has been associated with clinically significant differences in the clearance rates of efavirenz. Individuals that are carriers of wild-type GG genotype have the shortest median plasma half-lifes of 23 h, but those with TT genotype have longer median half-lifes of 48 h.

The maximum emtricitabine plasma concentrations of 1.5 – 2.0 mg/l occurs at 1 – 2 h post dose, with minimum concentrations of 0.04 – 0.09 mg/l. Bioavailability is high at 93%, with < 4% protein bound. Emtricitabine is primarily eliminated via the kidneys, with an oral clearance of 20 l/h. The AUC_{0-24} is 7 – 10 mg/h/l. The plasma t½ is 7 – 10 h, with an intracellular t½ of the triphosphate active metabolite of 39 h.

The tenofovir plasma Cmax is 0.3 – 0.4 mg/l and achieved at 2 – 3 h post dose. Minimum concentrations are 0.06 mg/l, with an AUC_{0-24} of 2 – 3 mg/h/l. Oral clearance is 35 – 40 l/h. Protein binding is low at < 0.7%, with 70 – 80% of systemically available tenofovir eliminated unchanged in urine. The plasma t½ of tenofovir is 12 – 17 h. The intracellular t½ of the tenofovir diphosphate is approximately > 60 h.

Special Populations
Race
Efavirenz: The pharmacokinetics of efavirenz appears to be similar among the racial groups.

Emtricitabine: No pharmacokinetic differences due to race have been identified following the administration of emtricitabine.

Tenofovir DF: There were insufficient numbers from racial and ethnic groups other than Caucasian to adequately determine the potential pharmacokinetic differences among these populations following the administration of tenofovir DF.

Gender
Efavirenz, Emtricitabine, and Tenofovir DF: Efavirenz, emtricitabine, and tenofovir DF pharmacokinetics are similar in male and female patients.

Pediatric and Geriatric Patients
is not recommended for pediatric administration. Pharmacokinetics of efavirenz, emtricitabine and tenofovir DF have not been fully evaluated in the elderly (>65 years of age).

Patients with Impaired Renal Function
In patients with creatinine clearance < 50 mL/min, emtricitabine and tenofovir exposure are increased, and a dose reduction is recommended by the manufacturer. In contrast, the elimination profile of efavirenz in these patients remains unchanged. For this reason, patients with creatinine clearance values < 50 mL/min should not receive TRUSTIVA.

Patients with Hepatic Impairment
Efavirenz: No significant effect on efavirenz pharmacokinetics was seen in subjects with mild hepatic impairment (Child-Pugh Class A) compared with controls. There were insufficient data to determine whether moderate or severe hepatic impairment (Child-Pugh Class B or C) affects efavirenz pharmacokinetics.

Emtricitabine: The pharmacokinetics of emtricitabine have not been studied in subjects with hepatic impairment; however, emtricitabine is not significantly metabolized by liver enzymes, so the impact of liver impairment should be limited.

Tenofovir DF: There were no substantial alterations in tenofovir DF pharmacokinetics in subjects with hepatic impairment compared with unimpaired subjects.

Indication
TRUSTIVA is indicated for the treatment of HIV-1 infection in adults.

Recommended Dose
Adults
The dose of TRUSTIVA is one tablet once daily taken orally on an empty stomach. Dosing at bedtime may improve the tolerability of nervous system symptoms.

Pediatrics
TRUSTIVA is not recommended for use in patients <18 years of age.

Renal Impairment
Because TRUSTIVA is a fixed-dose combination, it should not be prescribed for patients requiring dosage adjustment such as those with moderate or severe renal impairment (creatinine clearance <50 mL/min).

Rifampicin Coadministration
When TRUSTIVA is administered with rifampicin to patients weighing 50 kg or more, an additional 200 mg/day of efavirenz is recommended.

Mode of Administration
Oral

Contraindication
TRUSTIVA is contraindicated in patients with previously demonstrated hypersensitivity to any of the components of the product.
Table 1: Drugs That Are Contraindicated or Not Recommended for Use with TRUSTIVA.

Drug Class: Drug Name	Clinical Comment
Antifungal: voriconazole	Efavirenz significantly decreases voriconazole plasma concentrations, and coadministration may decrease the therapeutic effectiveness of voriconazole. Also, voriconazole significantly increases efavirenz plasma concentrations, which may increase the risk of efavirenz-associated side effects. Because TRUSTIVA is a fixed-dose combination product, the dose of efavirenz cannot be altered.
Antimigraine: Ergot derivatives (dihydroergotamine, ergonovine, ergotamine, methylergonovine)	Potential for serious and/or life-threatening reactions such as acute ergot toxicity characterized by peripheral vasospasm and ischemia of the extremities and other tissues.
Benzodiazepines: midazolam, triazolam	Potential for serious and/or life-threatening reactions such as prolonged or increased sedation or respiratory depression.
Calcium channel blocker: bepridil	Potential for serious and/or life-threatening reactions such as cardiac arrhythmias.
GI motility agent: cisapride	Potential for serious and/or life-threatening reactions such as cardiac arrhythmias.
Neuroleptic: pimozide	Potential for serious and/or life-threatening reactions such as cardiac arrhythmias.
St. John's wort (Hypericum perforatum)	May lead to loss of virologic response and possible resistance to efavirenz or to the class of nonnucleoside reverse transcriptase inhibitors (NNRTIs).

Warning and Precautions
Efavirenz:
Qtc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Interaction with Other Medicaments). Consider alternatives to TRUSTIVA when coadministered with a drug with a known risk of Torsade de Pointes or when administered to patients at higher risk of Torsade de Pointes.

General
TRUSTIVA should be taken cautiously in patients with bone, kidney or liver problems in anamnesis as well as in patients with psychiatric disorders and addictions.

Lactic Acidosis/Severe Hepatomegaly with Steatosis
Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of nucleoside analogs including tenofovir DF, a component of TRUSTIVA, in combination with other antiretrovirals. Particular caution should be exercised when administering nucleoside analogs to any patient with known risk factors for liver disease; however, cases have also been reported in patients with no known risk factors. Treatment with TRUSTIVA should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or pronounced hepatotoxicity (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

Patients Coinfected with HIV-1 and HBV
It is recommended that all patients with HIV-1 be tested for the presence of chronic HBV before initiating antiretroviral therapy. Patients who are coinfected with HIV-1 and HBV should be closely monitored with both clinical and laboratory follow up for at least several months after stopping treatment with TRUSTIVA. If appropriate, initiation of anti-hepatitis B therapy may be warranted. TRUSTIVA should not be administered with adefovir dipivoxil.

Coadministration with Related Products
Related drugs not for coadministration with TRUSTIVA include emtricitabine/rilpivirine/tenofovir DF, emtricitabine, tenofovir DF, and emtricitabine/tenofovir DF, which contain the same active components as TRUSTIVA. Efavirenz should not be coadministered with TRUSTIVA unless needed for dose-adjustment (e.g. with rifampicin). Due to similarities between emtricitabine and lamivudine, TRUSTIVA should not be coadministered with drugs containing lamivudine, including lamivudine/zidovudine, or lamivudine, abacavir sulfate/lamivudine, or abacavir sulfate/lamivudine/zidovudine.

Psychiatric Symptoms
Serious psychiatric adverse experiences have been reported in patients treated with efavirenz. Specific serious psychiatric events observed were severe depression, suicidal ideation, nonfatal suicide attempts, aggressive behavior, paranoid reactions, and manic reactions. Patients with serious psychiatric adverse experiences should seek immediate medical evaluation to assess the possibility that the symptoms may be related to the use of efavirenz and, if so, to determine whether the risks of continued therapy outweigh the benefits.

Nervous System Symptoms
Central nervous system symptoms were reported to have association with the treatment of efavirenz. These symptoms included dizziness, insomnia, impaired concentration, somnolence, abnormal dreams, and hallucinations. Other reported symptoms were euphoria, confusion, agitation, amnesia, stupor, abnormal thinking, and depersonalization. Patients should be informed that these common symptoms were likely to improve with continued therapy and were not predictive of subsequent onset of the less frequent psychiatric symptoms. Dosing at bedtime may improve the tolerability of these nervous system symptoms. Patients receiving TRUSTIVA should be alerted to the potential for additive central nervous system effects when TRUSTIVA is used concomitantly with alcohol or psychoactive drugs. Patients who experience central nervous system symptoms such as dizziness, impaired concentration, and/or drowsiness should avoid potentially hazardous tasks such as driving or operating machinery.
Late-onset neurotoxicity, including ataxia and encephalopathy (impaired consciousness, confusion, psychomotor slowing, psychosis, delirium), may occur months to years after beginning efavirenz therapy. Some events of late-onset neurotoxicity have occurred in patients with CYP2B6 genetic polymorphisms, which are associated with increased efavirenz levels despite standard dosing of TRUSTIVA. Patients presenting with signs and symptoms of serious neurologic adverse experiences should be evaluated promptly to assess the possibility that these events may be related to efavirenz use, and whether discontinuation of TRUSTIVA is warranted.

New Onset or Worsening Renal Impairment
Emtricitabine and tenofovir are principally eliminated by the kidney; however, efavirenz is not. Since TRUSTIVA is a combination product and the dose of the individual components cannot be altered, patients with creatinine clearance below 50 mL/min should not receive TRUSTIVA. Renal impairment, including cases of acute renal failure and Fanconi syndrome has been reported with the use of tenofovir disoproxil fumarate. Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients at risk for renal impairment, including patients who have previously experienced renal events while receiving adefovir dipivoxil. TRUSTIVA should be avoided with concurrent or recent use of a nephrotoxic agent.

Rash
Rash associated with blistering, moist desquamation, or ulceration are seen when treated with efavirenz. TRUSTIVA can be reinitiated in patients interrupting therapy because of rash. TRUSTIVA should be discontinued in patients developing severe rash associated with blistering, desquamation, mucosal involvement, or fever. Appropriate antihistamines and/or corticosteroids may improve the tolerability and hasten the resolution of rash.

Hepatotoxicity
Monitoring of liver enzymes before and during treatment is recommended for patients with underlying hepatic disease, including hepatitis B or C infection; patients with marked transaminase elevations; and patients treated with other medications associated with liver toxicity. Liver enzyme monitoring should also be considered for patients without pre-existing hepatic dysfunction or other risk factors.

Decreases in Bone Mineral Density
Assessment of bone mineral density (BMD) should be considered for patients who have a history of pathologic bone fracture or other risk factors for osteoporosis or bone loss. Although the effect of supplementation with calcium and vitamin D was not studied, such supplementation may be beneficial for all patients. If bone abnormalities are suspected, then appropriate consultation should be obtained.

Convulsions
Convulsions have been observed in patients receiving efavirenz, generally in the presence of known medical history of seizures. Caution must be taken in any patient with a history of seizures. Patients who are receiving concomitant anticonvulsant medications primarily metabolized by the liver, such as phenytoin and phenobarbital, may require periodic monitoring of plasma levels.

Immune Reconstitution Syndrome
Immune reconstitution syndrome has been reported in patients treated with combination antiretroviral therapy, including the components of TRUSTIVA. During the initial phase of combination antiretroviral treatment, patients whose immune system responds to such treatment may develop an inflammatory response to indolent or residual opportunistic infections (such as *Mycobacterium avium* infection, cytomegalovirus, *Pneumocystis jirovecii* pneumonia [PCP], or tuberculosis), which may necessitate further evaluation and treatment. Autoimmune disorders (such as Graves' disease, polymyositis, and Guillain-Barré syndrome) have also been reported to occur in the setting of immune reconstitution, however, the time to onset is more variable, and can occur many months after initiation of treatment.

Fat Redistribution
Redistribution/accumulation of body fat, including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting, breast enlargement and “cushingoid appearance” have been observed in patients receiving antiretroviral therapy. The mechanism and long-term consequences of these events are currently unknown. A causal relationship has not been established.

Pediatric Use
TRUSTIVA is not recommended for patients less than 18 years of age because it is a fixed-dose combination tablet containing a component, tenofovir disoproxil fumarate, for which safety and efficacy have not been established in this age group.

Geriatric Use
Clinical trials of efavirenz, emtricitabine, or tenofovir disoproxil fumarate did not include sufficient numbers of subjects aged 65 years and over to determine whether they respond differently from younger subjects. In general, dose selection for the elderly patients should be cautious, keeping in mind the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy.

Hepatic Impairment
TRUSTIVA is not recommended for patients with moderate or severe hepatic impairment because there are insufficient data.

Renal Impairment
Because **TRUSTIVA** is a fixed-dose combination, it should not be prescribed for patients requiring dosage adjustment such as those with moderate or severe renal impairment (creatinine clearance <50 mL/min).

Interaction with other Medicaments

QT Prolonging Drugs

There is limited information available on the potential for a pharmacodynamic interaction between TRUSTIVA and drugs that prolong the QTc interval. QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Warnings and Precautions). Consider alternatives to TRUSTIVA when coadministered with a drug with a known risk of Torsade de Pointes.
Drug Interactions
Efavirenz has been shown to induce CYP3A and CYP2B6. Other compounds that are substrates of CYP3A or CYP2B6 may have decreased plasma concentrations when coadministered with efavirenz. Coadministration of efavirenz with drugs primarily metabolized by these isozymes may result in altered plasma concentrations of the coadministered drug. Therefore, appropriate dose adjustments may be necessary for these drugs. Drugs that induce CYP3A activity (e.g., phenobarbital, rifampicin, rifabutin) would be expected to increase the clearance of efavirenz resulting in lowered plasma concentrations.

Since emtricitabine and tenofovir are primarily eliminated by the kidneys, coadministration of TRUSTIVA with drugs that reduce renal function or compete for active tubular secretion may increase serum concentrations of emtricitabine, tenofovir, and/or other renally eliminated drugs. Some examples include, but are not limited to, acyclovir, adefovir dipivoxil, cidofovir, ganciclovir, valacyclovir, and valganciclovir. Coadministration of tenofovir DF and didanosine should be undertaken with caution and patients receiving this combination should be monitored closely for didanosine-associated adverse reactions. Didanosine should be discontinued in patients who develop didanosine-associated adverse reactions. Suppression of $CD4^+$ cell counts has been observed in patients receiving tenofovir DF with didanosine 400 mg daily.

Lopinavir/ritonavir has been shown to increase tenofovir concentrations. Patients receiving lopinavir/ritonavir with TRUSTIVA should be monitored for tenofovir-associated adverse reactions. TRUSTIVA should be discontinued in patients who develop tenofovir-associated adverse reactions. Coadministration of atazanavir with TRUSTIVA is not recommended since coadministration of atazanavir with either efavirenz or tenofovir DF has been shown to decrease plasma concentrations of atazanavir. Also, atazanavir has been shown to increase tenofovir concentrations.

TRUSTIVA should not be administered concurrently with midazolam, triazolam, pimozide, bepridil, terfenadine, astemizole, cisapride, or ergot alkaloids because competition for CYP3A4 liver enzymes by Efavirenz what could result in inhibition of metabolism of these drugs and could be responsible for serious adverse events, including cardiac arrhythmias, prolonged or increased sedation and respiratory depression. TRUSTIVA also should not be co-administered with voriconazole, because Efavirenz significantly decreases its plasma concentrations. Drug should not be used during pregnancy unless clearly necessary, moreover pregnancy should be avoided in women receiving TRUSTIVA. Effective contraception should always be used (moreover because of the long half-life of Efavirenz, it is recommended for 12 weeks after discontinuation of treatment).

Size: 200x450mm
Pharmacode: Front - 815 & Back- 816
No.of colors: 01, Black
Note: Pharma code position and Orientation are tentative, will be changed based on folding size.

Nonprinting colors:
Diecut

Table 2: Established and other potentially significant drug interactions

Concomitant Drug Class: Drug Name	Effect	Clinical Comment
Antiretroviral Agents		
Protease inhibitor: Atazanavir	↓ atazanavir concentration ↑ tenofovir concentration	Co-administration of atazanavir with **TRUSTIVA** is not recommended. Co-administration of atazanavir with either efavirenz or tenofovir disoproxil fumarate decreases plasma concentrations of atazanavir. The combined effect of efavirenz plus tenofovir DF on atazanavir plasma concentrations is not known. Also, atazanavir has been shown to increase tenofovir disoproxil fumarate concentrations. There are insufficient data to support dosing recommendations for atazanavir or atazanavir/ritonavir in combination with **TRUSTIVA**.
Protease inhibitor: Fosamprenavir calcium	↓ amprenavir concentration	*Fosamprenavir (unboosted):* Appropriate doses of fosamprenavir and TRUSTIVA with respect to safety and efficacy have not been established. *Fosamprenavir/ritonavir:* An additional 100 mg/day (300 mg total) of ritonavir is recommended when TRUSTIVA is administered with fosamprenavir/ritonavir once daily. No change in the ritonavir dose is required when TRUSTIVA is administered with fosamprenavir plus ritonavir twice daily.
Protease inhibitor: Indinavir	↓ indinavir concentration	The optimal dose of indinavir, when given in combination with efavirenz, is not known. Increasing the indinavir dose to 1000 mg every 8 hours does not compensate for the increased indinavir metabolism due to efavirenz.
Protease inhibitor: Lopinavir/ritonavir	↓ lopinavir concentration ↑ tenofovir concentration	A dose increase of lopinavir/ritonavir to 600/150 mg (3 tablets) may be considered when used in combination with efavirenz in treatment-experienced patients where decreased susceptibility to lopinavir is clinically suspected (by treatment history or laboratory evidence). **Patients should be monitored for tenofovir-associated adverse reactions.** TRUSTIVA **should be discontinued in patients who develop tenofovir DF-associated adverse reactions**.
Protease inhibitor: Ritonavir	↑ ritonavir concentration ↑ efavirenz concentration	When ritonavir 500 mg every 12 hours was coadministered with efavirenz 600 mg once daily, combination was associated with a higher frequency of adverse clinical experiences (eg, dizziness, nausea, paresthesia) and laboratory abnormalities (elevated liver enzymes). Monitoring of liver enzymes is recommended when TRUSTIVA is used in combination with ritonavir.
Protease inhibitor: Saquinavir	↓ saquinavir concentration	Should not be used as sole protease inhibitor in combination with TRUSTIVA.
CCR5 co-receptor antagonist: Maraviroc	↓ maraviroc concentration	Efavirenz decreases plasma concentrations of maraviroc. Refer to the full prescribing information for maraviroc for guidance on coadministration with TRUSTIVA.
NRTI: Didanosine	↑ didanosine concentration	Higher didanosine concentrations could potentiate didanosine-associated adverse reactions, including pancreatitis and neuropathy. **In adults weighing >60 kg, the didanosine dose should be reduced to 250 mg if coadministered with** TRUSTIVA. **Data are not available to recommend a dose adjustment of didanosine for patients weighing <60 kg. Coadministration of** TRUSTIVA **and didanosine should be undertaken with caution and patients receiving this combination should be monitored closely for didanosine-associated adverse reactions. For additional information, please consult the didanosine prescribing information.**
Other Agents		
Anticoagulant: Warfarin	↑ or ↓ warfarin concentration	Plasma concentrations and effects potentially increased or decreased by efavirenz.
Anticonvulsants: Carbamazepine	↓ carbamazepine concentration ↓ efavirenz concentration	There are insufficient data to make a dose recommendation for TRUSTIVA. Alternative anticonvulsant treatment should be used.
Phenytoin Phenobarbital	↓ anticonvulsant concentration ↓ efavirenz concentration	Potential for reduction in anticonvulsant and/or efavirenz plasma levels; periodic monitoring of anticonvulsant plasma levels should be conducted.
Antidepressants: Bupropion	↓ bupropion concentration	The effect of efavirenz on bupropion exposure is thought to be due to the induction of bupropion metabolism. Increases in bupropion dosage should be guided by clinical response, but the maximum recommended dose of bupropion should not be exceeded.
Sertraline	↓ sertraline concentration	Increases in sertraline dose should be guided by clinical response.
Antifungals: Itraconazole Ketoconazole Posaconazole	↓ itraconazole concentration ↓ hydroxyitraconazole concentration ↓ ketoconazole concentration ↓ posaconazole concentration	Since no dose recommendation for itraconazole can be made, alternative antifungal treatment should be considered. Drug interaction trials with TRUSTIVA and ketoconazole have not been conducted. Efavirenz has the potential to decrease plasma concentrations of ketoconazole. Avoid concomitant use unless the benefit outweighs the risks.
Anti-infective: Clarithromycin	↓ clarithromycin concentration ↑ 14-OH metabolite concentration	Clinical significance unknown. No dose adjustment of TRUSTIVA is recommended when given with clarithromycin. Alternatives to clarithromycin, such as azithromycin, should be considered. Other macrolide antibiotics, such as erythromycin, have not been studied in combination with TRUSTIVA.
Antimycobacterial: Rifabutin	↓ rifabutin concentration	Increase daily dose of rifabutin by 50%. Consider doubling the rifabutin dose in regimens where rifabutin is given 2 or 3 times a week
Antimycobacterial: Rifampin	↓ efavirenz concentration	If TRUSTIVA is coadministered with rifampicin to patients weighing 50 kg or more, an additional 200 mg/day of efavirenz is recommended.
Calcium channel blockers: Diltiazem	↓ diltiazem concentration ↓ desacetyl diltiazem concentration ↓ N-monodesmethyl diltiazem concentration	Diltiazem dose adjustments should be guided by clinical response (refer to the prescribing information for diltiazem). No dose adjustment of TRUSTIVA is necessary when administered with diltiazem.
HMG-CoA reductase inhibitors: Atorvastatin Pravastatin Simvastatin	↓ atorvastatin concentration ↓ pravastatin concentration ↓ simvastatin concentration	Plasma concentrations of atorvastatin, pravastatin and simvastatin decreased. Consult the complete prescribing information for the HMG-CoA reductase inhibitor for guidance on individualizing the dose.
Immunosuppressants: Cyclosporine, tacrolimus, sirolimus, and others metabolized by CYP3A	↓ immuno-suppressant	Decreased exposure of the immunosuppressant may be expected due to CYP3A induction by efavirenz. These immunosuppressants are not anticipated to affect exposure of efavirenz. Dose adjustments of the immunosuppressant may be required. Close monitoring of immunosuppressant concentrations for at least 2 weeks (until stable concentrations are reached) is recommended when starting or stopping treatment with TRUSTIVA.
Narcotic analgesic: Methadone	↓ methadone concentration	Co-administration of efavirenz in HIV-1-infected individuals with a history of injection drug use resulted in decreased plasma levels of methadone and signs of opiate withdrawal. Methadone dose was increased by a mean of 22% to alleviate withdrawal symptoms. Patients should be monitored for signs of withdrawal and their methadone dose increased as required to alleviate withdrawal symptoms.

[a] This table is not all-inclusive.

Pregnancy and Lactation
Pregnancy Category D:
Efavirenz may cause fetal harm when administered during the first trimester to a pregnant woman. Pregnancy should be avoided in women receiving TRUSTIVA. Barrier contraception must always be used in combination with other methods of contraception (e.g., oral or other hormonal contraceptives). Because of the long half-life of efavirenz, use of adequate contraceptive measures for 12 weeks after discontinuation of TRUSTIVA is recommended. Women of childbearing potential should undergo pregnancy testing before initiation of TRUSTIVA. If this drug is used during the first trimester of pregnancy, or if the patient becomes pregnant while taking this drug, the patient should be apprised of the potential harm to the fetus. TRUSTIVA should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus, such as in pregnant women without other therapeutic options.

Lactation
The Centers for Disease Control and Prevention recommend that HIV-1-infected mothers not breastfeed their infants to avoid risking postnatal transmission of HIV-1. It is not known whether efavirenz, tenofovir or emtricitabine are excreted in human milk. Because of both the potential for HIV-1 transmission and the potential for serious adverse reactions in nursing infants, **mothers should be instructed not to breast-feed if they are receiving** TRUSTIVA.

Side Effects
The most significant adverse reactions linked with Efavirenz use are CNS and psychiatric symptoms as headache, disturbance in attention, abnormal dreams, somnolence, insomnia, depression, and anxiety. However, dosing at bedtime may improve the tolerability of nervous system symptoms. Moreover, these psychiatric and CNS side effects related to efavirenz usually begin during the first one or two days of therapy and generally resolve after the first two to four weeks. The other possible side effects associated with Tenofovir disoproxil fumarate are kidney function decline or failure and bone abnormalities. Renal function (creatinine clearance and serum phosphate) should be monitored every 4 weeks during the first year and then every 3 months. TRUSTIVA is not recommended for patients with moderate or severe renal impairment (creatinine clearance <50 ml/min).

Postmarketing experiences: encephalopathy

Symptoms and Treatment of Overdose
If overdose occurs, the patient should be monitored for evidence of toxicity, including monitoring of vital signs and observation of the patient's clinical status; standard supportive treatment should then be applied as necessary.

Administration of activated charcoal may be used to aid removal of unabsorbed efavirenz.

Hemodialysis can remove both emtricitabine and tenofovir disoproxil fumarate (refer to detailed information below), but is unlikely to significantly remove efavirenz from the blood.

Efavirenz
Increased nervous system symptoms and involuntary muscle contractions have been reported.

Emtricitabine
Limited clinical experience is available at doses higher than the therapeutic dose of emtricitabine. No severe adverse reactions were reported with single doses of emtricitabine 1200 mg.

It is not known whether emtricitabine can be removed by peritoneal dialysis.

Tenofovir DF
Limited clinical experience at doses higher than the therapeutic dose of tenofovir disoproxil fumarate 300 mg is available. The effects of higher doses are not known.

Tenofovir DF is efficiently removed by hemodialysis with an extraction coefficient of approximately 54%. Following a single 300 mg dose of tenofovir disoproxil fumarate, a 4-hour hemodialysis session removed approximately 10% of the administered tenofovir disoproxil fumarate dose.

Storage Condition
Store at below 30°C and protect from moisture.

Shelf Life
24 months

Dosage forms and packaging available
30 Tablets in HDPE container

Manufacturer
HETERO LABS LIMITED
Unit V, Sy. No 439,440,441 & 458
TSIIC Formulation SEZ, Polepally Village,
Jadcherla Mandal, Mahaboobnagar Dist.
Telangana, India

Product Registration Holder:
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